



## Kabul

One of the few cities with a fully functional Wi-Max infrastructure



## Dell Studio

Dell has jumped into the bandwagon, its Studio lineup is now available with WiMax capability

## WiMax devices for businesses

Soma enterprises, has tied up with BSNL and will offer WiMax enabled devices for Business customers in India

## Feature

dividing each channel into sub-channel groups. Insofar that the channel assigned to any user will be optimized based on which will perform best for the current user's location, environment and equipment.

In the original 802.16-2004 specification, OFDM-256 was used, which meant each channel would be composed of 256 sub-carriers and multiple user access available via TDMA (or FDMA). The 802.16e-2005 spec adds OFDMA 128/512/1024/2048, while the old OFDM-256 is kept for compatibility reasons.

What a difference an alphabet can make!

### SOFDMA

SOFDMA is OFDMA but with scalability to enable its usage in a wider variety of scenarios, especially mobile. While the original 802.16-2004 standard was meant to enable wireless internet access to fixed locations such as homes and businesses, where a WiMAX receiver would be affixed outside a the building similar to a dish antenna, in 802.16e-2005 many specifications were added which would enable a mobile system to access the network.

SOFDMA is scalable in that it allows for changing the bandwidth allocation to channels from 1.25 MHz to 20 MHz. Smaller channels such as those to be used by mobile devices are optimized for lesser system complexity, while higher bandwidth channels such as those used by fixed point subscribers at home or in offices, are optimized for higher bandwidth. Even the modulation scheme is scalable such that a different encoding scheme can be depending on the quality of the signal.

### WIMAX VS LTE

There is another war of standards coming up! LTE stands for Long Term Evolution, and it widely thought of as an upgrade path from the current GSM 3G / 3.5G standards. It has backing from a majority of the GSM providers, but unlike WiMAX it is not based on an open standard.

Although LTE cannot compete with WiMAX in all its facilities, in the mobile sphere they are both quite well matched. In fact they both use the same OFDM and MIMO technologies

for communication. LTE has the upper hand in speeds right now, with the soon to come 802.16m standard, WiMAX may well overtake LTE.

Unlike LTE for WiMAX mobile access is already available, and already starting to get implemented. As LTE cannot compete with WiMAX in all fronts, it is unlikely that LTE will replace WiMAX, but as far as the mobile space is concerned, the battle is ON!

### HOW WiMAX WORKS



### Other 802.16e features

Many other features were also added in 802.16e-2005, to improve its mobility characteristics, an important one being MIMO (multiple-in multiple-out). Other than the mobility enhancements, there were also the addition of HARQ (Hybrid Automatic Repeat Request), and QoS (Quality of Service) improvements.

### MIMO

To improve the range and speed characteristics, multiple antennae can be used at the transmitting and receiving station. This improves the maximum range of the transmission and the data throughput. Using this in WiMAX means that without increasing the output power of the transmitter, and while keeping the same bandwidth, they will be able to provide better performance to a longer range. This is of obvious benefit in

mobile access, as it enables customers to get better access with wider area coverage, all without significant cost increase to the broadcaster.

### QoS

Networks are used for a wide variety of communication, from internet gaming to bank transactions, each type of communication requiring different network characteristics. While we may consider out internet surfing quite important, there are many other protocols which are less resilient to network outages or interruptions. VoIP for example requires a constant guaranteed minimum rate of data flow for communication to persist.

To guarantee that such time and delay critical communications are not interrupted, QoS protocols are employed to give them higher priority. Communications of different classes are given different priority. The QoS in 802.16e-2005 divides network communication into 5 classes:

- 1.Unsolicited Grant Service
- 2.Extended Real-time Polling Service
- 3.Real-time Polling Service
- 4.Non-real-time Polling Service
- 5.Best Effort

VoIP falls in the second class, and our favorite HTTP communication is sadly last, given only as much bandwidth as is left over in the end.

### Conclusion

Are we ready for internet as easily available as telephone signals? I believe we are, we have been good, and we deserve our candy. WiMAX is not the technology of the future; it is already something that is being implemented in many areas around the globe, a good example being WiBro, which is an 802.16 implementation in South Korea.

802.16 holds a lot of promise for more than just internet, it is potentially a technology that can provide a wireless backbone for the telecommunications industry. Our homes are already starting to get wireless, WiMAX aims to make entire cities wireless, which if you live in India you can clearly sympathize with.

No more broken connection due to wiring faults, or overloaded trucks passing through narrow streets! Just like mobile phones have made the telecommunication an indispensable tool, and taken us from one landline per home to one phone per person, so does WiMAX hold the potential to someday make internet as personal and as available.

8th ANNIVERSARY  
CARNIVAL

digit contests digit DEALS digit events

**DON'T MISS** all the excitement this month.  
**WIN, LEARN, DISCUSS, STRIKE DEALS** and much more

Visit us at **thinkdigit.com**

world are shelling out big money for WiMax. Gastonia paid up \$40,000